TARANEH-ASR: SCALING PERSIAN SPEECH RECOGNITION AND FORCED ALIGNMENT ACROSS SINGING AND CONVERSATIONAL DOMAINS

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ABSTRACT

Automatic Speech Recognition (ASR) and Forced Alignment (FA) in low-resource languages such as Persian remain severely hindered by a scarcity of diverse, domain-adapted corpora, especially in acoustically complex registers like singing voice and informal broadcast speech. In this paper, we present a unified framework spanning open data curation, foundation model fine-tuning, standardized evaluation, and alignment readout. We introduce two open datasets and benchmarks for ASR consisting of the 102-hour human-annotated *Persian Free Music*, the first copyright-free singing corpus, and *Persian Media Speech*, a 617-hour semi-supervised broadcast and conversational dataset. We fine-tune and release *Taraneh*, a family of five models spanning Whisper and Qwen3-ASR backbones (Taraneh-WBase, -WMedium, -WLarge, -Q0.6B, -Q1.7B) on ≈1,560 hours of composite speech, establishing new benchmarks across seven test sets, including an 18.00% WER on PSRB as an out-of-domain evaluation set. Furthermore, we show that internal cross- and self-attention geometry can be extracted without retraining to perform word-level FA. With gold-guided head selection and silence trimming, *Taraneh-WMedium* achieves 37.8 ms MAE on FarsDat, outperforming off-the-shelf neural aligners. All corpora, benchmarks, and models are publicly released.

Index Terms— Persian ASR, forced alignment, low-resource speech, singing voice recognition

1. INTRODUCTION

Automatic Speech Recognition (ASR) has achieved impressive accuracy in high-resource languages through large-scale multilingual pre-training [1–3]. However, for low-resource languages such as Persian (Farsi), spoken by over 110 million people across diverse dialects [4], open speech resources remain scarce. Historically, early Persian speech initiatives such as FARSDAT [5] provided acoustic-phonetic read sentences, while contemporary public benchmarks like Mozilla Common Voice [6] and FLEURS [7] broadened evaluation. Nevertheless, these corpora feature almost exclusively clean, read speech. Acoustically complex and expressive domains, specifically singing voice with rich musical accompaniment and colloquial, spontaneous social media speech, remain completely unaddressed in open Persian resources, causing foundational speech models to degrade. Sung speech in particular has been studied in some languages like English, through karaoke and polyphonic corpora such as DSing and DALI [8, 9], with no Persian counterpart.

To address data scarcity, self-training and pseudo-labeling effectively adapt multilingual foundation models to low-resource targets [10, 11]. While Whisper [1] and Qwen3-ASR [3] possess rich cross-lingual representations, zero-shot Persian accuracy degrades on out-of-domain speech. Compounding this, Persian orthography exhibits severe surface variability, optional short vowels, arbitrary Zero-Width Non-Joiner (ZWNJ) usage, and loose clitic detachment, which inflates Word Error Rate (WER) unless resolved by standardized text normalization.

Concurrently, downstream tasks such as subtitling and pronunciation scoring demand precise word boundaries. Traditional Forced Alignment (FA) relies on GMM-HMMs like the Montreal Forced Aligner (MFA) [12], requiring language-specific lexicons. While Connectionist Temporal Classification (CTC) [13–15], pipeline aligners [16], and slot-filling aligners (e.g., Qwen3-ForcedAligner [3, 17]) provide neural alternatives, they require separate acoustic models prone to domain mismatch. Although pre-trained cross-attentions contain word alignments [18], whether attention alignment survives domain fine-tuning across encoder-decoder and decoder-only LLM-ASR architectures on non-Latin scripts remains unstudied.

We present a unified framework addressing both challenges by expanding open Persian speech domains, standardizing orthography, and unlocking dual ASR and forced alignment within fine-tuned foundation models. Our primary contributions are:

1. **Persian Free Music Dataset & Benchmark (102 h):** The first open copyright-free singing corpus (1,700 tracks across six genres) with human-generated text, containing 102 h of music for training alongside a held-out 2,000-utterance benchmark evaluated under musical accompaniment.
2. **Persian Media Speech Dataset & Benchmark (617 h):** A 617-hour corpus collected from Persian news channels on Telegram together with entertainment clips from Instagram and TikTok, covering diverse speaking styles across a wide variety of environmental and acoustic noise conditions, alongside a curated 1,185-utterance evaluation benchmark.
3. **Five Released Models (*Taraneh* Family):** We fine-tune and release *Taraneh* across five models: *Taraneh-WBase*, *-WMedium*, *-WLarge* (Whisper backbones), and *-Q0.6B*, *-Q1.7B* (Qwen3-ASR backbones) on ≈1,560 h of composite speech, reporting results across seven benchmarks.
4. **Internal Attention Forced Alignment (Dual ASR+FA):** A zero-training attention readout recipe for fine-tuned checkpoints. Gold-guided head selection and silence trimming enable *Taraneh-WMedium* to achieve 37.8 ms Mean Absolute Error (MAE) on FarsDat, outperforming dedicated neural aligners.

2. PROPOSED CORPORA AND BENCHMARKS

2.1. Persian Free Music Dataset

To address the scarcity of open Persian musical speech resources, we collected 1,700 copyright-free tracks spanning six genres (Pop,

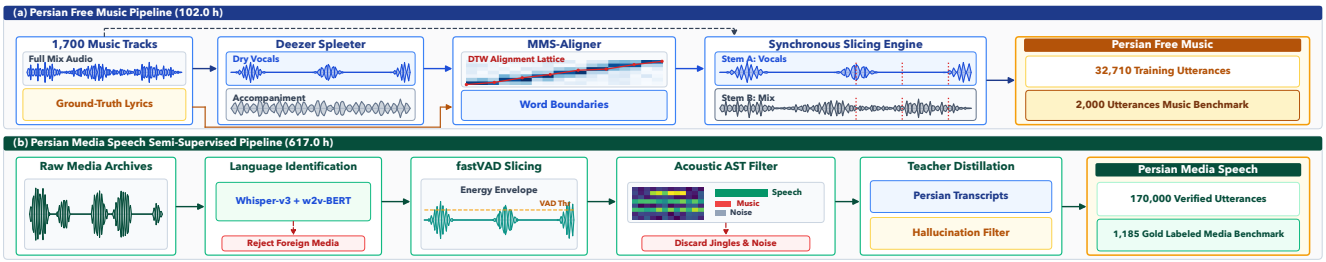


Fig. 1. Overview of proposed dataset construction workflows: (a) Persian Free Music pipeline combining Deezer Spleeter source separation with MMS forced alignment to slice full-mix tracks; (b) Persian Media Speech semi-supervised distillation pipeline combining Whisper LID, fastVAD, fine-tuned 3-class AST filtering, and teacher pseudo-labeling.

Table 1. Statistics of training mixture and proposed held-out evaluation benchmarks. Both eval sets are strictly disjoint from training.

Corpus	Hours	Utterances	Labels	Domain
Training Data Mixture				
Common Voice 20 (fa)	369.3	329,019	Native	Read speech
Persian Media Speech	617.0	170,000	Pseudo-labeled	Conversational / broadcast
Farsi YouTube	306.9	127,308	Native	Spontaneous / narrated
Persian Music	204.0	69,962	Native	Singing (mix + stem)
VoxLingua107 (fa)	51.5	19,344	Pseudo-labeled	In-the-wild (YouTube)
FLEURS (fa)	11.5	2,994	Native	Read speech
Total Training Mixture	$\approx 1{,}560$	**718,627**		
Proposed Evaluation Benchmarks				
Music	3.4	2,000	Native	Singing (mix)
Media	4.1	1,185	Native	Conversational / broadcast

Rap, Rock, Electronic, Traditional Iranian, R&B) providing 102 h of music for training alongside a held-out 2,000-utterance evaluation benchmark, paired with human-verified ground-truth lyrics.

Directly aligning multi-instrumental songs yields severe boundary degradation due to accompaniment masking. As shown in Fig. 1(a), we resolve this via a two-stream separation and alignment workflow: (1) *Vocal Separation & Conditioning:* Deezer Spleeter (2-stem model) [19] isolates dry vocal stems from the instrumental mix, which are resampled to 16 kHz mono WAV, peak-normalized, and passed through an energy gate to suppress residual accompaniment leakage; (2) *Lyric Forced Alignment:* MMS-based FA [15] aligns conditioned vocals against human lyrics, extracting word-level boundary timestamps; (3) *Dynamic Dual-Stem Slicing:* Word boundaries dynamically slice both isolated dry vocal stems and original full-mix tracks into matched segments (mean 7.50 s); and (4) *Partitioning & Dual-Stem Training:* A held-out benchmark of 2,000 utterances (≈3.4 h) evaluates singing ASR under accompaniment, containing exclusively vocals mixed with musical accompaniment. For model training, we have 102 h of unique singing data; crucially, we include both the full musical mix and the dry vocal stem during training, yielding 204 h of dual-stem training speech (69,962 clips, Table 1) to ensure acoustic robustness to both isolated singing and dense instrumental accompaniment.

2.2. Persian Media Speech Corpus

Capturing informal, conversational, and broadcast registers requires scaling beyond read speech. We harvested uncurated multimedia archives from high-volume Persian news channels on public Telegram communities together with entertainment clips from Instagram and TikTok, and filtered them via a 5-stage distillation pipeline (Fig. 1(b)): (1) *Language Identification (LID):* Audio streams are first evaluated with the LID head of Whisper Large-v3 [1] to eliminate non-Persian languages, followed by a secondary verification pass using a W2V-BERT-based language identification model [20] to ensure authentic Persian speech; (2) *VAD Slicing:* Continuous audio is segmented into 3–20 s speech chunks (mean 11.87 s) using the fastVAD engine [21], stripping extended pauses; (3) *Acoustic Filtering:* We fine-tuned an Audio Spectrogram Transformer (AST) [22] on AudioSet [23] as a 3-class classifier (speech, music, noise), discarding non-speech and music-corrupted segments; (4) *Teacher Pseudo-Labeling:* Clean speech chunks are transcribed using our fine-tuned Whisper-LargeV3 teacher checkpoint (fine-tuned on all other datasets presented in Table 1) with beam search decoding; and (5) *Quality Control:* Hallucinations and token loops are pruned using character-per-second bounds ($2 \leq \text{cps} \leq 25$) and repetition penalties.

2.3. Training Mixture and Proposed Evaluation Sets

We combine native labels (Common Voice, FLEURS, YouTube captions, lyric-aligned music) with pseudo-labels (Persian Media Speech, VoxLingua107 [24]). To ensure acoustic robustness to narrow-band transmission channels without relying on external telephony training speech, we applied a telephony effect (simulating 8 kHz bandpass filtering and μ-law codec compression) as dynamic data augmentation to 10% of the training utterances.

3. INTERNAL ATTENTION FORCED ALIGNMENT

Because sequence-to-sequence speech decoding inherently requires mapping acoustic frames to linguistic units over time, foundation speech decoders implicitly preserve fine-grained temporal-acoustic alignments within their internal attention geometry [18]. We leverage this property to enable fine-tuned ASR checkpoints to perform word-level forced alignment without training separate aligners, achieving unified, simultaneous ASR and FA.

Given acoustic input $\mathbf{X}$ and normalized transcript $\mathbf{y}$, the audio encoder yields temporal frame representations at 50 Hz (T_{enc} frames, 20 ms stride). Under teacher-forcing, the autoregressive decoder is conditioned on standard task-prompt prefix tokens followed by character-tokenized text $\mathbf{y} = [c_1, \ldots, c_T]$, where whitespace delimiters are explicitly retained as independent tokens. We extract the cross-attention tensor $\mathbf{A} \in [0,1]^{L \times H \times T \times T_{\text{enc}}}$ corresponding to lexical token positions $\mathbf{y}$ (excluding prefix prompt steps) across all L layers and H attention heads.

To suppress high-frequency frame jitter, we apply a 1D median filter ($w = 3$) along the acoustic dimension. For a designated subset of specialized alignment heads $\mathcal{H} \subset \{1, \ldots, L\} \times \{1, \ldots, H\}$ isolated via gold-guided selection (detailed below), each head slice

Table 2. Performance evaluation (WER / CER %) of zero-shot baselines, ablations, published external systems, and proposed fine-tuned models across seven Persian benchmarks after unified text normalization. Best performance per column among fine-tuned models is highlighted in bold.

Model	Training Data	FLEURS	CommonVoice	YouTube	Music (Ours)	Media (Ours)	PSRB (Unseen)	VisualEars (Unseen)
Pre-trained Baselines (Zero-Shot)								
Whisper-LargeV3 (PT)	-	19.97 / 6.60	37.04 / 12.80	35.44 / 18.66	58.44 / 28.10	88.48 / 54.27	–	–
Qwen3-ASR 1.7B (PT)	-	30.40 / 9.31	53.40 / 24.70	49.06 / 23.72	82.90 / 49.35	45.92 / 21.88	–	–
Dataset Ablation Models								
Whisper-Medium	Existing Data (5 Sets)	7.33 / 2.40	4.42 / 1.90	15.76 / 8.60	46.10 / 23.20	29.67 / 18.42	–	–
Whisper-Medium	Existing + Music (204h)	6.80 / 2.20	4.63 / 2.10	15.71 / 8.50	24.47 / 11.40	24.86 / 13.65	–	–
Proposed Released Models (*Taraneh* Family, trained on all ≈1,560 h)								
Taraneh-WBase (Whisper-Base)	All Data	7.88 / 2.93	10.76 / 4.45	19.02 / 10.30	37.69 / 17.67	24.43 / 13.46	26.88 / 12.26	11.58 / 3.97
Taraneh-WMedium (Whisper-Med)	All Data	4.16 / 1.52	3.47 / 1.67	14.29 / 8.26	**22.79 / 10.92**	**14.97 / 8.99**	19.14 / 8.73	6.23 / 1.89
Taraneh-WLarge (Whisper-L-v3)	All Data	**3.86 / 1.33**	**3.34 / 1.53**	**13.42 / 7.99**	22.98 / 11.00	16.06 / 9.54	**18.00 / 8.17**	**5.31 / 1.56**
Taraneh-Q0.6B (Qwen3-0.6B)	All Data	7.11 / 2.54	9.65 / 3.86	18.87 / 10.12	35.97 / 15.83	22.32 / 11.73	24.61 / 10.67	9.44 / 3.12
Taraneh-Q1.7B (Qwen3-1.7B)	All Data	6.25 / 2.15	7.85 / 3.13	18.21 / 9.91	34.50 / 15.05	21.91 / 11.46	22.79 / 9.98	8.05 / 2.64
External Benchmark Comparison								
Best Prior System*	–	4.45 / 2.35	19.50 / 4.18	14.72 / 7.61	–	–	19.30 / 8.75	3.68 / 1.26

*Best prior published system per benchmark: FLEURS (Gemini-3.5 Flash) [25]; Common Voice (Gooya-1.4) [26]; YouTube (Gooya-1.4) [26]; PSRB (Avanegar) [27]; VisualEars (Gemini-3.5 Flash) [25, 28].

is column ℓ_2-normalized to balance inter-head energy before ensembling:

$$\bar{\mathbf{A}}(t_{\text{dec}}, t_{\text{enc}}) = \frac{1}{|\mathcal{H}|} \sum_{(l,h)\in\mathcal{H}} \frac{\mathbf{A}_{l,h}(t_{\text{dec}}, t_{\text{enc}})}{\|\mathbf{A}_{l,h}(:, t_{\text{enc}})\|_2}. \quad (1)$$

Dynamic Time Warping (DTW) [29] on cost matrix $\mathbf{C} = -\bar{\mathbf{A}}$ with standard monotonic step constraints yields the optimal acoustic-to-token path. The continuous temporal path is then projected into discrete word boundaries $[t_{\text{start}}, t_{\text{end}}]$ by segmenting character spans at explicit whitespace positions.

Decoder-Only Readout: For decoder-only Qwen3-ASR, acoustic and linguistic tokens are processed jointly within a causal self-attention stream. Using forward hooks on thinker layers, we extract attention from text query positions to interleaved audio key tokens (downsampled to ∼80 ms frame resolution). Applying median filtering and DTW path search on this cross-modal sub-matrix yields word boundaries.

Gold-Guided Head Selection and Silence Trimming: Prior zero-shot alignment recipes [18] selected heads using unsupervised entropy or diagonal concentration heuristics. However, we observe that downstream fine-tuning fundamentally shifts internal representations: certain heads specialize as "attention sinks" [30] that absorb residual probability mass or fixate degenerately on punctuation and padding frames. Unsupervised metrics prioritize these sinks, causing DTW alignment to collapse (boundary MAE > 100 ms). To resolve this, we introduce gold-guided head selection: using a small tuning pool of 300 clips (∼15 min of speech, strictly disjoint from evaluation splits), we isolate each candidate head (l, h), execute per-head DTW, and rank all $L \times H$ heads by empirical boundary MAE against ground-truth human annotations.

Empirical alignment capability concentrates sharply in a sparse subset of deep transformer layers: Taraneh-WMedium (L19/H8, L16/H12, $k = 2$), Taraneh-WLarge (L23/H3, L26/H7, $k = 2$), and Taraneh-Q0.6B and Q1.7B (L22/H7, L24/H12, L20/H15, $k = 3$). Ensembling only these top-k heads reduces MAE from 108.3 ms to 37.8 ms. Additionally, an energy-based silence trimmer (-35 dB relative frame threshold over 20 ms windows) clamps boundary tokens to speech onset and offset, eliminating systematic edge pinning.

FarsDat Gold Benchmark and Cross-Fitting: To establish a rigorous Persian forced-alignment benchmark, we construct an out-of-domain evaluation testbed from FarsDat-small [5] (excluded from ASR training), featuring studio speech with human phonetic and word time boundaries. Because original FarsDat lexicons provide only romanized phonemic labels, we recover canonical Persian orthography by transcribing utterances with fine-tuned Whisper, computing space-insensitive multi-speaker consensus per sentence ID (sid), and verifying token correspondences via dynamic programming phonetic-skeleton matching. The resulting curated benchmark comprises 6,039 verified utterances totaling 36,634 gold word intervals (73,268 boundary events). To eliminate data snooping, all evaluations employ disjoint 5-fold cross-fitted head selection ($sid \pmod 5$), ensuring no evaluation clip is aligned using heads selected on its own sentence partition.

Table 3. Forced alignment evaluation on FarsDat gold benchmark (36.6k words) with cross-fitted heads and silence trimming.

Alignment System	MAE (ms)	Median (ms)	≤50ms (%)	≤100ms (%)	IoU
Internal Attention Forced Alignment (*Taraneh* Models)					
Taraneh-WMedium (2 heads + trim)	**37.8**	**21.6**	**80.7**	94.4	**0.837**
Taraneh-WLarge (2 heads + trim)	38.1	22.9	80.4	**94.5**	0.836
Taraneh-Q0.6B (3 heads + trim)	47.2	33.4	67.7	92.7	0.791
Taraneh-Q1.7B (3 heads + trim)	48.5	35.5	65.7	92.6	0.781
Off-the-Shelf Neural Aligners					
ctc-forced-aligner (MMS-300m) [14]	63.7	43.8	56.6	84.5	0.702
Qwen3-ForcedAligner-0.6B (official) [3]	90.2	36.6	61.3	77.1	0.651
Whisper-Medium FT (unsup. top-10) [18]	108.3	64.8	40.4	67.0	0.646
Lexicon-Assisted Dedicated Aligner (Reference)					
Montreal Forced Aligner 3.4 (MFA) [12]*	20.6	11.4	93.9	98.8	0.904

*MFA is trained in-domain with FarsDat gold phonemic pronunciations (held-out MAE 21.3 ms), serving as an acoustic upper bound.

4. EXPERIMENTAL SETUP

We fine-tune and release five *Taraneh* speech models on the ≈1,560 h mixture (Table 1). All five models are trained via AdamW [31] with cosine decay. ASR is evaluated across seven benchmarks: standard sets (Common Voice, FLEURS, YouTube), our proposed held-out sets (Music 2,000 utterances, Media 1,185 utterances, Table 1), alongside unseen out-of-domain benchmarks: PSRB [27] and VisualEars [28]. For Forced Alignment, checkpoints are evaluated on the 6,039-clip FarsDat gold benchmark against: (1) ctc-forced-aligner [14]; (2) Qwen3-ForcedAligner-0.6B [3]; and (3) in-domain MFA [12]. Metrics include word-boundary Mean Absolute Error (MAE $= \frac{1}{2N}\sum_{i=1}^{N}[|\hat{s}_i - s_i| + |\hat{e}_i - e_i|]$ over predicted $[\hat{s}_i, \hat{e}_i]$ and gold $[s_i, e_i]$ word intervals), median error, accuracy (≤50 ms, ≤100 ms), and mean Intersection-over-Union (IoU).

4.1. Inference-Time Persian Text Normalization Pipeline

In automatic speech recognition, rigorous text normalization during evaluation is a foundational prerequisite for standardized benchmarking, ensuring that WER reflects genuine acoustic misrecognition rather than superficial orthographic variations. Due to Perso-Arabic script characteristics, Persian exhibits pronounced surface variability: conflicting Unicode glyph encodings across Arabic and Persian keyboards (e.g., *Kaf* and *Yeh*), optional short vowel diacritics (*harakat*), arbitrary ZWNJ usage in compound words, and rampant colloquial spoken reductions. Left unaddressed, these superficial mismatches artificially inflate WER by 15–30% despite correct acoustic decoding [4, 27, 32].

Crucially, our fine-tuned models are trained directly on natural orthographic transcripts without synthetic alteration; text normalization is executed strictly at inference time, applied symmetrically to both reference transcripts and decoded hypotheses prior to WER and CER computation.

We designed a deterministic, four-stage scoring normalization pipeline: (1) *Numeral Verbalization & Spoken Normalization:* Indo-Arabic and Latin numerals or decimals are expanded into spoken Persian words via hierarchical scale decomposition (10^3 *hezār*, 10^6 *milyun*, decimal *momayyez*), resolving digit-word mismatches. (2) *Unicode Canonicalization:* Over 40 Arabic/Persian glyph variants and ligatures (e.g., *Kaf*, *Yeh*, *Ta-Marbuta*, *Hamza*) are mapped to canonical Persian Unicode. ASCII and Persian punctuation, short vowel diacritics (*tanwin*, *kasra*, *damma*, *fatha*), and ZWNJ characters are stripped; (3) *Morphological Clitic Detachment:* Compound spacing is standardized, and a 3-pass morphological rule engine detaches bound enclitics, pronominals (*-am*, *-at*, *-ash*, *-mān*, *-tān*, *-shān*), superlatives (*-tarin*), and verb plurals (*-ānd*, *-īd*, *-īm*), while joining continuous verbal prefixes (*mi-*, *nemi-*); (4) *Pair-Wise Tokenization Disambiguation:* During scoring, if hypothesis and reference share identical non-whitespace characters but differ in spacing, hypothesis tokenization is aligned to reference, ensuring WER penalizes acoustic recognition errors rather than arbitrary segmentation conventions.

5. RESULTS AND DISCUSSION

5.1. Speech Recognition Performance

Table 2 details ASR results. Zero-shot foundation models struggle severely: Whisper-LargeV3 yields 37.04% WER on Common Voice, 58.44% on Music, and 88.48% on Media, while Qwen3-ASR-1.7B obtains 53.40% on Common Voice and 82.90% on Music. In-domain training proves decisive: adding our 204-hour dual-stem *Persian Free Music* corpus reduces singing WER from 46.10% to 24.47%. Adding the 617-hour *Persian Media Speech* corpus cuts Media WER from 29.67% to 14.97%. Among released models, Taraneh-WLarge achieves state-of-the-art accuracy: 3.34% WER (1.53% CER) on Common Voice, 3.86% on FLEURS, and 13.42% on YouTube-Farsi. Taraneh-WMedium achieves 22.79% on Music and 14.97% on Media. Unseen evaluations verify robust transfer: 18.00% WER on PSRB [27] and 5.31% on VisualEars [28].

Comparison with Prior Systems: Taraneh-WLarge demonstrates substantial gains over published Persian systems: outperforming Gooya-1.4 [26] on Common Voice (3.34% vs. 19.50% WER), FLEURS (3.86% vs. 11.08%), and YouTube (13.42% vs. 14.72%); surpassing open Shenava Koochik (3.86% vs. 7.12%) [33] and commercial Gemini-3.5 Flash (3.86% vs. 4.45%) [25]; achieving 5.31% on unseen VisualEars [28] (near Koochik's 5.23%); and attaining 18.00% WER on PSRB [27], surpassing both open Faster-Whisper [34] (33.93%) and the top commercial system Avanegar (19.30%).

5.2. Forced Alignment Performance

Table 3 reports forced alignment evaluations on the 6,039-clip FarsDat gold benchmark. Attention readout from fine-tuned Taraneh-WMedium yields 37.8 ms MAE and 21.6 ms median error, with 80.7% of word boundaries localized within 50 ms of human annotations (0.837 mean IoU). Taraneh-WLarge demonstrates comparable alignment precision (38.1 ms MAE, 80.4% within 50 ms), with 94.4% of predicted boundaries falling within 100 ms.

Relative to dedicated off-the-shelf neural aligners, attention readout reduces MAE by 25.9 ms compared to MMS CTC (63.7 ms) and by 52.4 ms compared to official Qwen3-ForcedAligner (90.2 ms). For decoder-only models, self-attention readout from Taraneh-Q0.6B yields 47.2 ms MAE (67.7% within 50 ms), consistent with its coarser $\sim$80 ms acoustic token framing. In-domain MFA attains 20.6 ms MAE; however, it relies on acoustic model training and human-annotated phonemic pronunciations, serving as an acoustic reference rather than an open-vocabulary text aligner.

5.3. Ablations and Diagnostic Findings

Head Selection & Stability: Unsupervised selection [18] collapses on fine-tuned models (108.3 ms MAE) due to attention sinks. Gold-guided ranking on 15 min of speech isolates deep alignment heads (L16/L19 for Taraneh-WMedium, L23/L26 for Taraneh-WLarge, L20–L25 for Taraneh-Q0.6B) with optimal $k \in \{2, 3\}$, showing 100% split stability on Taraneh-WLarge.

Readout vs. Slot-Filling: Fine-tuning Qwen3-ASR with explicit timestamp slots (67.1 ms MAE) outperforms official Qwen3-FA (90.2 ms) but underperforms direct attention readout (47.2 ms), confirming that internal attention better preserves continuous alignment geometry.

Granularity, Softmax, & Trimming: Character attention matches uniform interpolation against gold phones (42.1 vs. 41.2 ms MAE), confirming word-level rather than phone-level resolution. Re-applying softmax degrades MAE (74.0 to 101.6 ms), whereas silence trimming (-35 dB) removes edge-pinning, improving boundary accuracy by >30 ms.

Limitations: Gold FA was evaluated on FarsDat studio speech; conversational alignment remains constrained by data scarcity, and attention readout targets words rather than phonemes.

6. CONCLUSION

We presented an open framework for Persian speech, releasing the first open 102-hour (204 h dual-stem) singing corpus, the 617-hour Persian Media Speech corpus, a text normalization pipeline, and the *Taraneh* family of five fine-tuned models trained on $\approx$1,560 h of composite speech. Our training-free attention recipe enables speech foundation models to simultaneously perform accurate ASR and broadcast-quality forced alignment (37.8 ms MAE), outperforming dedicated aligners. By publicly releasing all checkpoints, datasets, and evaluation pipelines, we establish a reproducible foundation to catalyze future research in low-resource speech recognition, expressive vocal processing, and unified alignment architectures. Future work will investigate extending internal attention readout to phoneme-level resolution and scaling conversational alignment to in-the-wild spontaneous speech.

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